

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

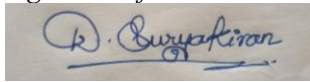
COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Code of Conduct:

*I D.Suryakiran (192325057) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer:

The neural network is a deep learning model trained to predict student scores. The training MSE is 0.85, while the testing MSE is 5.20. This large gap indicates that the model has poor generalization and is most likely overfitting.

Analysis:

- During training, the deep neural network learns the training data very well, resulting in a low MSE (0.85).
- During testing, the high MSE (5.20) shows that the model cannot accurately predict unseen student data.
- This happens when the model memorizes the training samples instead of learning general patterns.

Possible Reasons:

1. Overfitting due to too many hidden layers or neurons.
2. Insufficient training data, causing the network to memorize examples.
3. Lack of regularization (Dropout or L2 Regularization).
4. Improper hyperparameter tuning, such as a high learning rate or too many training epochs.
5. Data distribution mismatch between training and testing datasets.

Corrective Measures:

- Use Dropout layers to reduce overfitting.
- Apply L2 Regularization (Weight Decay).
- Use Early Stopping to stop training before overfitting occurs.
- Increase the size and diversity of the training dataset or use data augmentation where applicable.
- Tune hyperparameters such as learning rate, batch size, number of epochs, and network architecture.
- Use cross-validation to improve model generalization.

Conclusion:

The significant increase in testing MSE compared to training MSE indicates that the deep learning model is overfitting. Techniques such as Dropout, L2 Regularization, Early Stopping, proper hyperparameter tuning, and increasing training data can improve the model's ability to generalize and reduce the testing error.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

